For a vector space V, a set of vectors in V is a <u>basis</u> for V if:

(1) The vectors generate/span V, i.e. they are a spanning set for V.
(2) The set of vectors is linearly independent.

Sort of a "minimal" spanning set.

We talked about spanning set before, the major new thing is linear independence, that guarantees no redundancy, makes it as efficient as possible.

Why? Well if set is linearly dependent, some vector is a linear combination of others so if we throw that vector out, the overall span stays the same, meaning that vector was redundant.

And if linearly independent, we can't throw any vector out as we definitely can't generate the full space then.

In fact we can't generate that thrown out vector for one!

So it's sort of optimal in that sense, generates full space, but does it without redundancy.

Basis is very important - when trying to work with a vector space, whatever you are trying to do, often the first thing to do to help understand it is to identify a basis for it.

Particularly if it has a finite basis, breaking down an infinite space into examining a finite basis often makes things much more accessible.

When we had a vector space/subspace, presumably with an infinite number of vectors, we broke that down into a finite spanning set (when possible) as that was a much more efficient way of describing the space.

So now we are shoring up that process itself to make the spanning set itself as efficient as possible. That's what basis is largely about, that efficiency.

An immediate question that we alluded to when talking about $\mathbb{R}^3$ was if a vector space has a finite basis, must any two bases have the same number of elements?

Shall investigate that later.

We often want to find a basis for a subspace. Keep in mind, subspace is a vector space in its own right so finding a basis for it isn't fundamentally a different question.

However, often times a vector space which is quite well known, say $\mathbb{R}^3$, some subspace of it might not have an obvious basis like the original space does (the standard vectors, $\{e_1, e_2, e_3\}$).

Amongst such subspaces are Nul A, Col A, Row A for some arbitrary m x n real matrix A. So trying to find a strategy for determining each of those is a good start.

For each of these we use the RREF to make this easier.

When we practically apply this, as usual REF will be enough, but for the purposes of establishing why it works, easier to use RREF.

(1) Nul A: As usual use nonpivot columns which we find via RREF. That gives us free variables so then we express the answer in parametric form. Those vectors multiplying the free variables are a basis for Nul A.

Why? Well we've already showed they are a spanning set, just need to show those vectors are always linearly independent. They are because each vector has its own free variable attached.

If one of them was linear combination of others, that term could be removed meaning we wouldn't have the number of free variables suggested - contradiction.

So those are linearly independent, meaning to find Nul A it's the usual approach of solving homogeneous system.

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(2) Col A: The pivot columns of A are a basis for Col A (note that's A, not its RREF!)

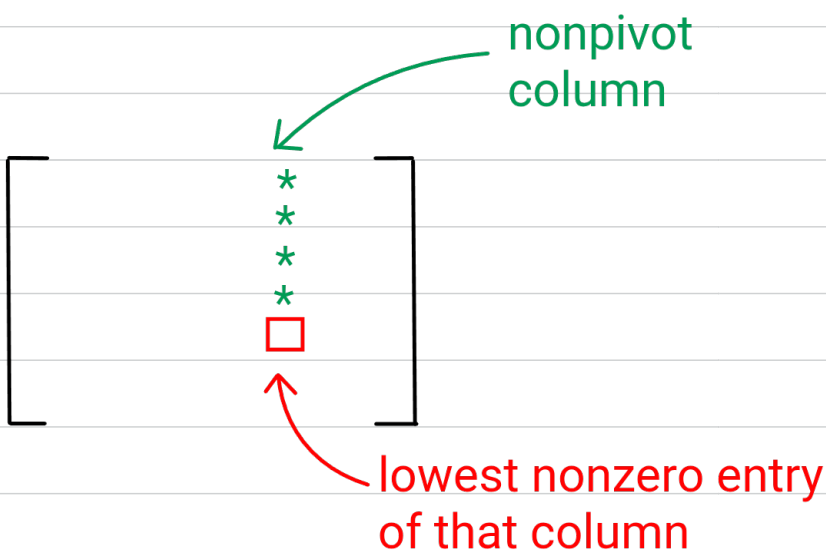
Why? Well a useful result for this and beyond is that if a set of nonzero vectors $\{v_1, v_2, \dots, v_r\}$ is linearly dependent then for some $1 \leq k \leq r$, v_k is a linear combination of $\{v_1, v_2, \dots, v_{k-1}\}$, i.e. some vector is a linear combination of the vectors prior to it. And this is natural because by the time we reach the last one if this was never true, the overall set would become linearly independent, a contradiction.

With this in mind why are pivot columns always a basis for Col A? Well in RREF each of those columns has a single entry which is 1, rest of the column has 0s. And those 1s are in different rows as no row can have multiple pivots meaning no pivot column can be a linear combination of the other pivot columns.

Thus in RREF quite clearly no pivot column is a linear combination of the other pivot columns so the set of pivot columns is indeed linearly independent.

The harder thing is to show each nonpivot column is a linear combination of the pivot columns. If we do that though, it will prove pivot columns are indeed a basis for Col A.

So take an arbitrary nonpivot column:



Consider all pivot columns to the left of this nonpivot column which have a pivot in the row of that lowest nonzero entry or in a row above it.

E.g.

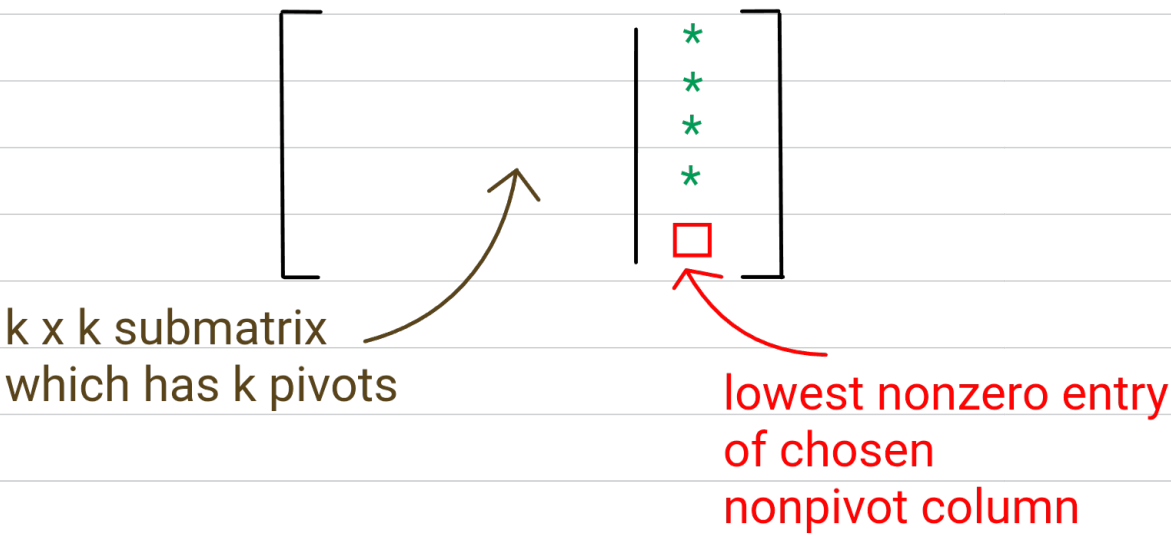
$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

For the nonpivot fourth column we would take the first two columns, not the third since the third column's pivot is in a lower row than the lowest nonzero entry, 3, in that fourth column.

It turns out those pivot columns always generate that nonpivot column.

Why? Say those pivot columns have k pivots. Then consider the $k \times k$ submatrix with the rows and columns of those k pivots, the lowest row by necessity being the row of the lowest nonzero entry in our chosen nonpivot column - that row must have a pivot since it does have at least one nonzero entry with the matrix being in RREF.

Thus look at the augmented matrix:



In this augmented matrix the coefficient matrix is $k \times k$ with k pivots so is invertible, guaranteeing the augmented column is a linear combination of those, columns, regardless of the exact entries.

Thus that nonpivot column is a linear combination of those pivot columns as desired (in the lower rows below that lowest nonzero entry all of those columns have only 0 entries since we only took pivot columns whose pivot was in that lowest row or above, and in the augmented column the box entry is the lowest nonzero entry.

So indeed pivot columns of A are basis for Col A. But again must use pivot columns of A, not a REF or its RREF. Why? Because row operations can change column space. For example:

$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

First column is only pivot column. But those two don't have same column space, first is $y = x$, second is $y = 0$.

(3) Row A: Fortunately row operations do not alter the row space like they can column space. So simply use the nonzero (pivot) rows in a REF or the RREF of A.

Since zero rows contribute nothing, those nonzero rows do generate the row space.

And those rows are linearly independent since in the RREF no such row can be a linear combination of the rows below it since the lower rows all have 0s in the column where the upper row has its leading entry.

Thus no nontrivial linear combination of those nonzero rows can be 0 as the highest row involved in such a combination would be linear combination of rows below it which we just said is not possible.

So nonzero rows in REF/RREF of A are basis for the row space.

So to sum it up for the three common subspaces we are often interested in an arbitrary real matrix A.

You put A in REF/RREF then proceed as follows:

(1) Nul A. Solve $Ax = 0$ via typical basic/free variables and then use the parametric vector form to obtain a basis. Namely the vectors which multiply the free variables in that form are the desired basis.

(2) Col A. Locate the pivot columns of A via the REF. Those pivot columns are a basis for Col A. Note that you must use the pivot columns of A, not the pivot columns of some REF of A as that may not give the same subspace!

(3) Row A. Use the nonzero rows in any REF of A, those will be a basis for Row A. Here, unlike Col A, we must use REF, not the original A as the pivot (top) rows of A often won't be a basis for Row A.

4.3.13 Want to find Nul A, Col A, Row A of

$$A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}$$

This is given to be row equivalent to:

$$B = \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Since B is in REF we find these as the pivots.

From this REF we immediately have for bases:

Col A: $\left\{ \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ -6 \\ 8 \end{bmatrix} \right\}$

Row A: $\left\{ \begin{bmatrix} 1 \\ 0 \\ 6 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 5 \\ 3 \end{bmatrix} \right\}$

From B, x_3, x_4 are free variables, and relevant equations for null space are:

$$2x_2 + 5x_3 + 3x_4 = 0$$
$$x_1 + 6x_3 + 5x_4 = 0$$

Thus

$$2x_2 = -5x_3 - 3x_4$$
$$x_1 = -6x_3 - 5x_4$$

So solution is (multiply all by two to remove fractions)

$$\begin{bmatrix} -12x_3 - 10x_4 \\ -5x_3 - 3x_4 \\ 2x_3 \\ 2x_4 \end{bmatrix} = x_3 \begin{bmatrix} -12 \\ -5 \\ 2 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -10 \\ -3 \\ 0 \\ 2 \end{bmatrix}$$

And therefore for the basis of Nul A we have:

Nul A: $$\left\{ \begin{bmatrix} -12 \\ -5 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -10 \\ -3 \\ 0 \\ 2 \end{bmatrix} \right\}$$

Even though matrix is not square so different number of rows and columns, with them also not in same space, yet they have same number of basis elements. Interesting question we will address later is whether that's always true.

This is end of material needed for Midterm 1.

4.4 Coordinates

Say vector space V has basis $\{v_1, v_2, \ldots, v_n\}$

So indirectly we are assuming has finite basis which of course isn't always true (e.g. $\mathbb{P}$, the space of polynomials over $\mathbb{R}$, as mentioned in worksheet).

In some contexts we work in, finite basis is required to make the situation workable for us. In others it's not such a big deal, context is huge.

After all if we accept the idea of working with an infinite basis, then writing a vector as a linear combination of those basis elements, it might become an infinite sum, i.e. infinite series!

And as you might recall from Calc II (e.g. Math 52), then it becomes an entirely different situation! Not something we are really equipped to deal with.

In this setting each $v \in V$ can be written uniquely in form:

$$v = c_1 v_1 + c_2 v_2 + \ldots + c_n v_n$$

Why? Well v being able to be written as such a a linear combination is immediate since that set is a basis and basis must span V by definition.

Why unique? Well say there was another representation

$$v = d_1 v_1 + d_2 v_2 + \ldots + d_n v_n$$

Then subtracting gives us

$$0 = (c_1 - d_1)v_1 + (c_2 - d_2)v_2 + \ldots + (c_n - d_n)v_n$$

But basis is linearly independent so this forces

$$c_1 = d_1, \; c_2 = d_2, \ldots, \; c_n = d_n$$

meaning the two representations were actually the same, proving the uniqueness.

Due to this uniqueness we define v's "coordinates" in this basis to be

$$(c_1, c_2, \ldots, c_n)$$

I.e. the scalars multiplying the basis elements in order.

When we say "order", there is some prescribed order we use for the basis elements. Not that using a different order would change the nature of the basis, but for cataloging purposes and to keep things consistent, we assign an order.

So each vector in V can be thought of as such a point, and any such point, set of scalars, corresponds to a unique point in V likewise as vector space is closed under addition and scalar multiplication so any such point will create an element in V as stated.

As such, each vector in V can be thought of, i.e. corresponds to, a unique point in $\mathbb{R}^n$

Adding two vectors in V is equivalent to adding their representations in $\mathbb{R}^n$ in the usual way so this works nicely.

More generally for $\mathcal{B}$ an n element basis for V (basis elements given in a specified order), $[x]_{\mathcal{B}}$

is the element of $\mathbb{R}^n$ whose entries are the coefficients of x in that basis.

With this, gives rise to $T: V \dashrightarrow \mathbb{R}^n$ defined by

$$T(x) = [x]_{\mathcal{B}}$$

which is a one-to-one linear transformation from V onto $\mathbb{R}^n$

Being one-to-one and onto is apparent from earlier ideas. One-to-one since two distinct vectors cannot have the same representation, onto due to vector space being closed under addition and scalar multiplication.

And for linear transformation:

$$T(cx + y) = [cx + y]_{\mathcal{B}}$$

$$= [cx]_{\mathcal{B}} + [y]_{\mathcal{B}}$$

$$= c[x]_{\mathcal{B}} + [y]_{\mathcal{B}}$$

$$= cT(x) + T(y)$$

proving linear transformation.

Via this, a vector space having a basis with n elements can be associated well with $\mathbb{R}^n$ which is very helpful since we can use what we know about $\mathbb{R}^n$ to draw conclusions about V, even if V looks very different structurally than $\mathbb{R}^n$.

That's in a way why studying $\mathbb{R}^n$ was so useful since any real vector space with a finite basis can be associated with some $\mathbb{R}^n$.

This also relates to another interesting issue that we we dealt with earlier in the course, the question of whether every linear transformation is a matrix transformation.

We showed when the linear transformation was from some $\mathbb{R}^n$ to some $\mathbb{R}^m$ then that was true.

And this association shows that it will always be true for any linear transformation from a real vector space V to another real vector space W, as long as both V and W both have a finite basis.

And it also hints as to why this is not true in every vector space setting, as if either V or W does not have a finite basis, this will break down. Logical because how could you even multiply a finite size matrix with a vector which has infinitely many "coordinates".

Hence why such linear transformations won't be matrix transformations.